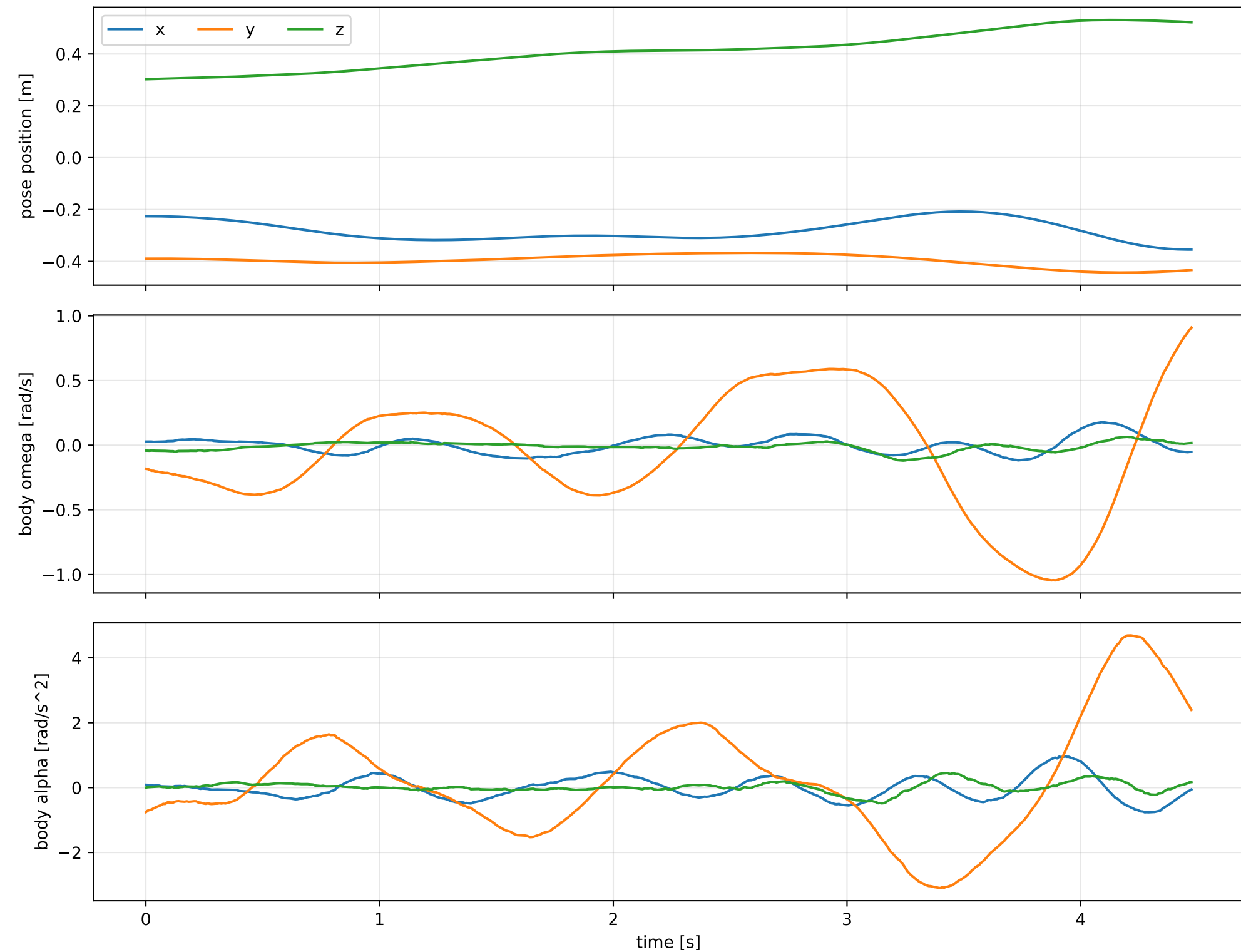
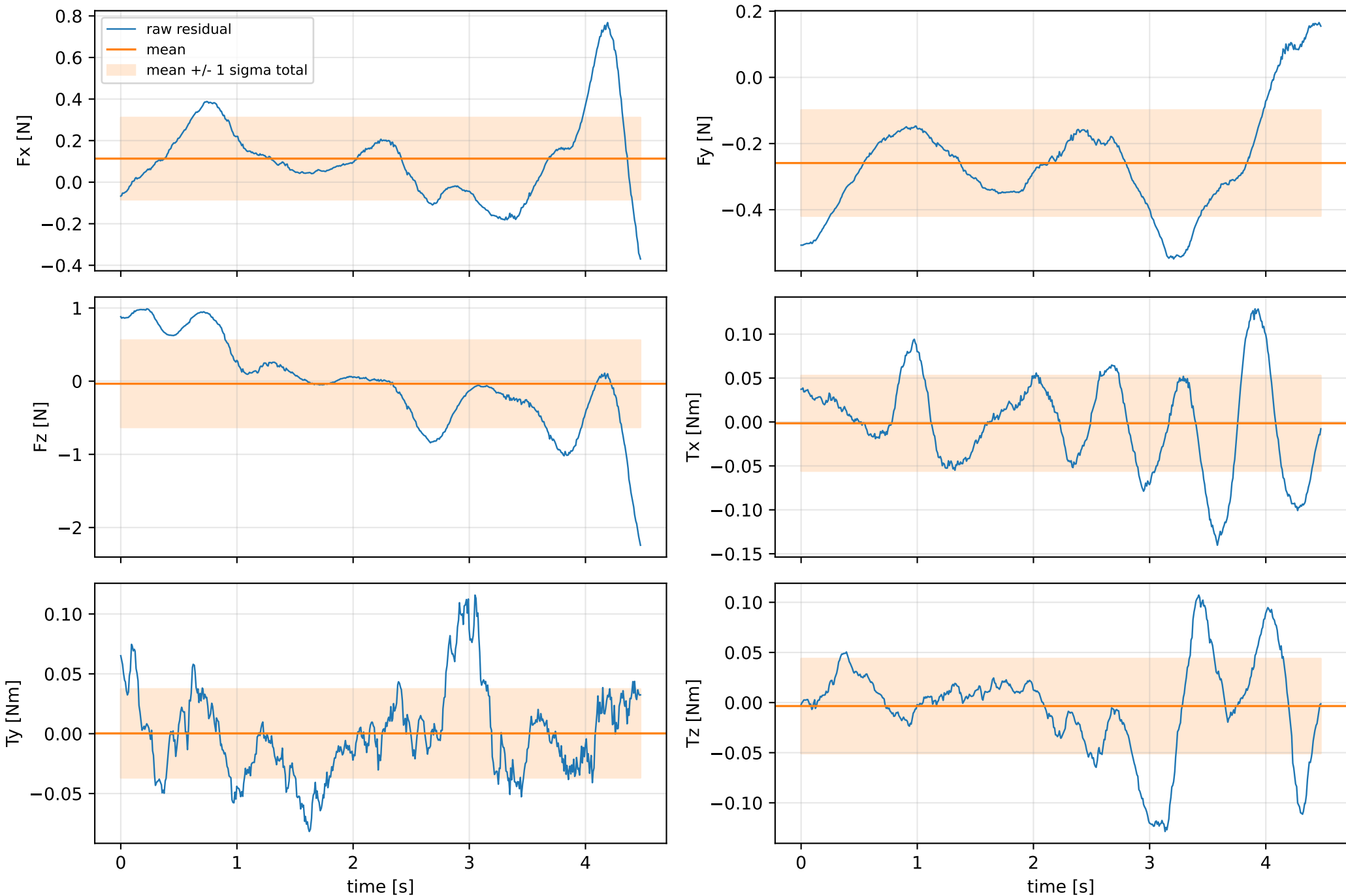


prior_free: SG trajectory and derived motion



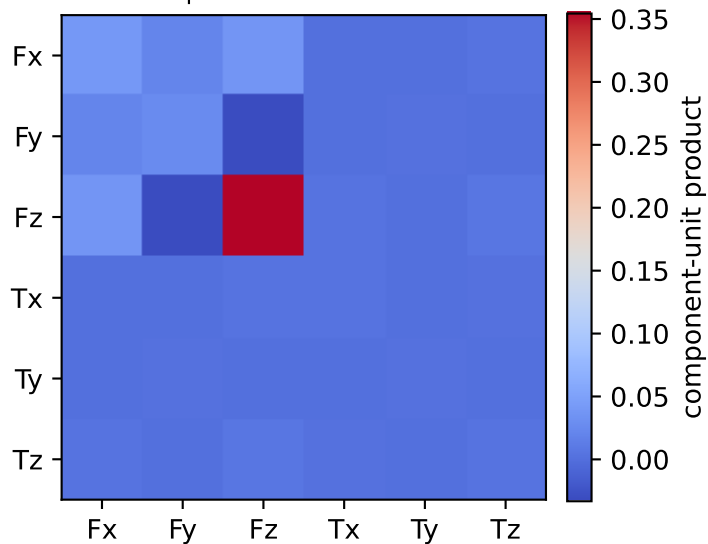
prior_free: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



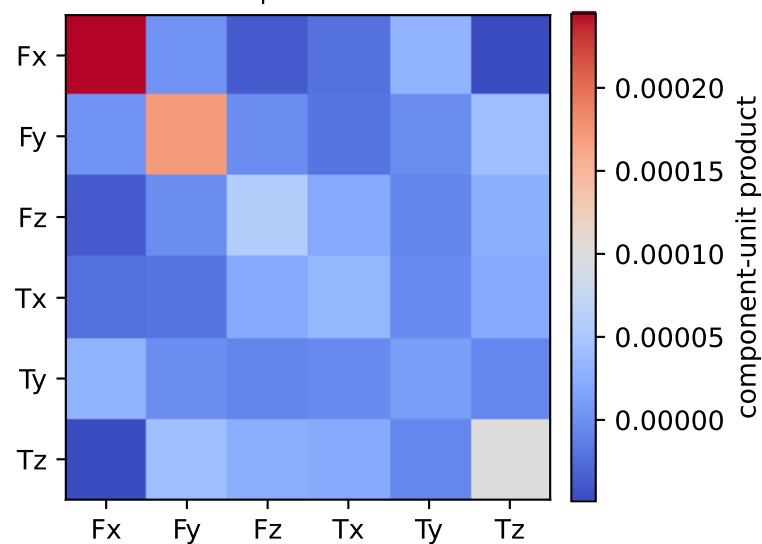
vector RMS about zero: force 0.70661 N, torque 0.080986 Nm; component std about mean: force [0.1974 0.16 0.5953], torque [0.0543 0.0371 0.0473]

prior_free: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.004 0.005 0.052 0.361]

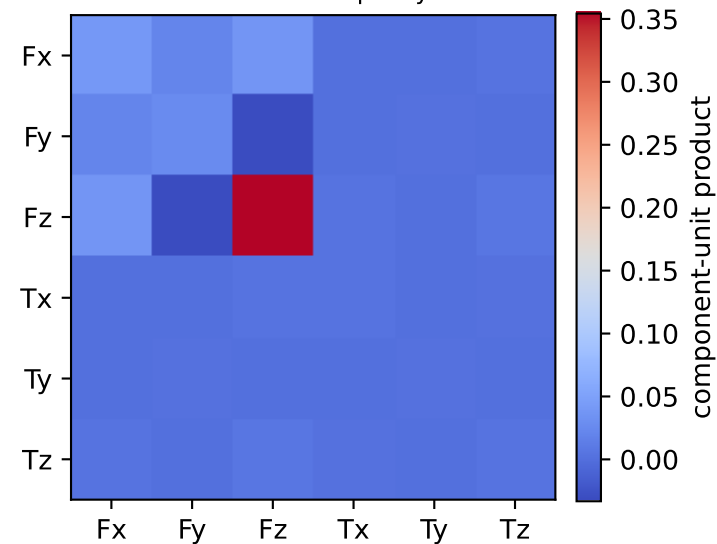
empirical total covariance



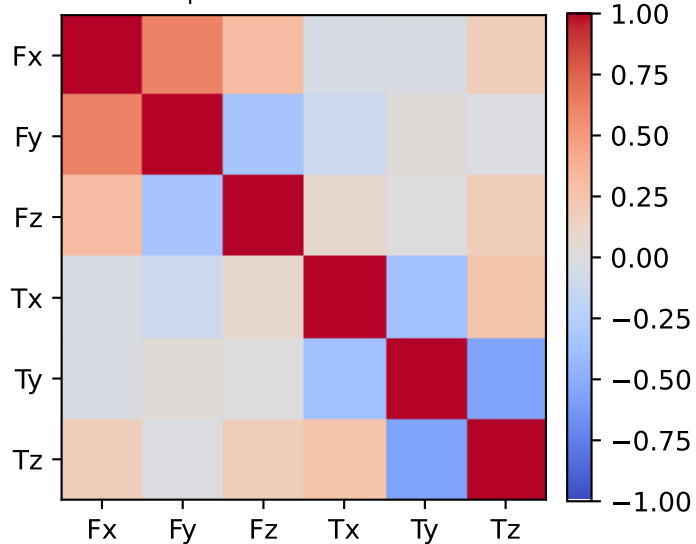
mean full-SG prediction covariance



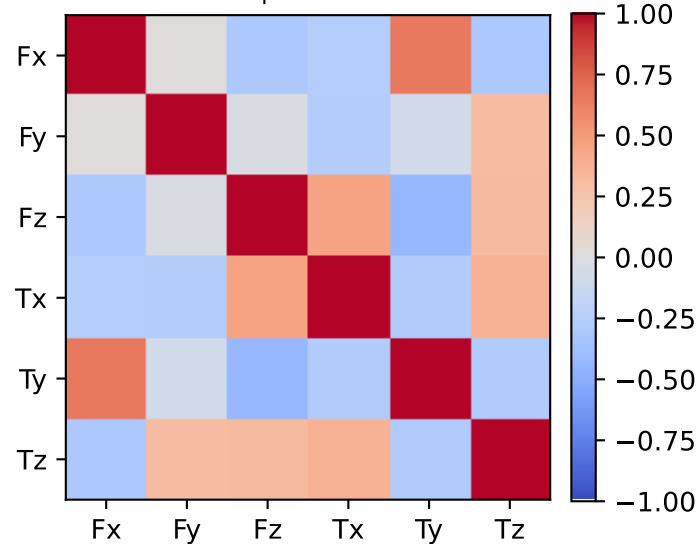
PSD excess model discrepancy covariance



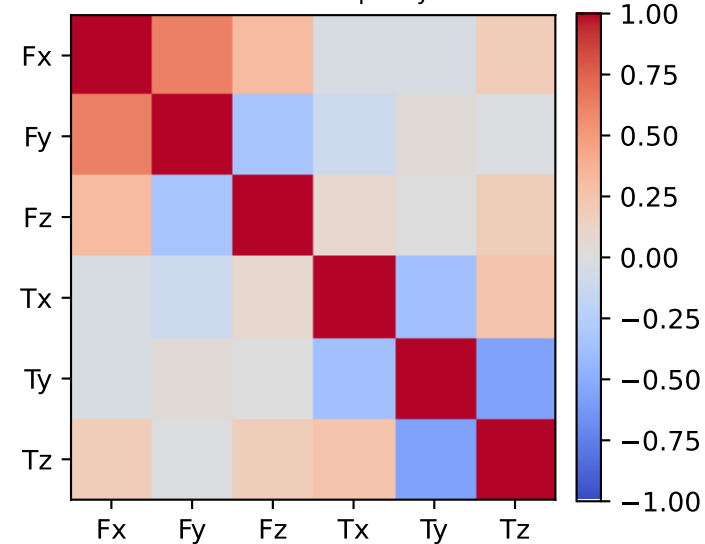
empirical total correlation



mean full-SG prediction correlation

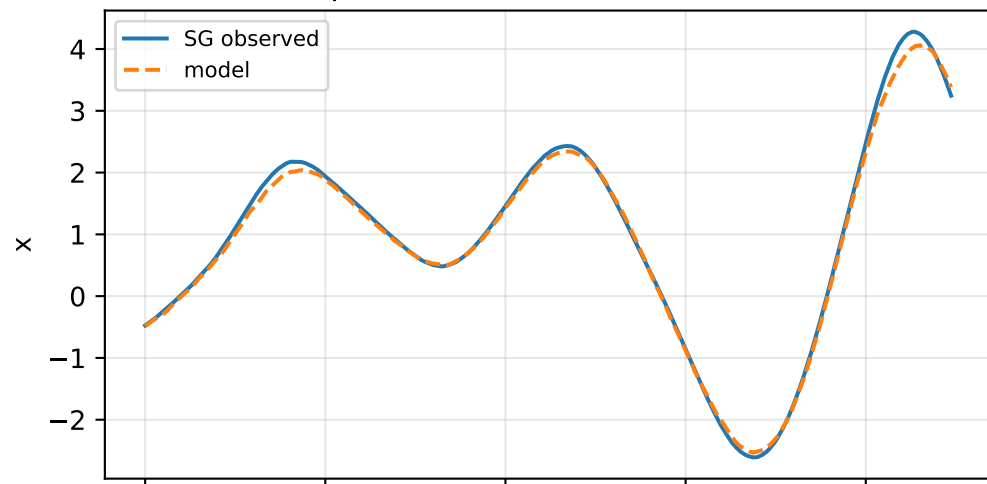


PSD excess model discrepancy correlation

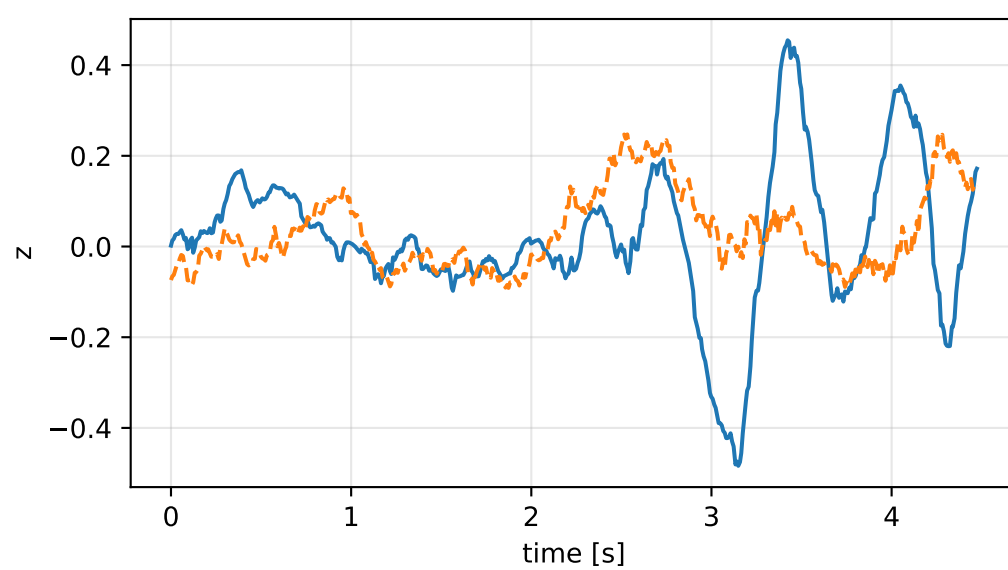
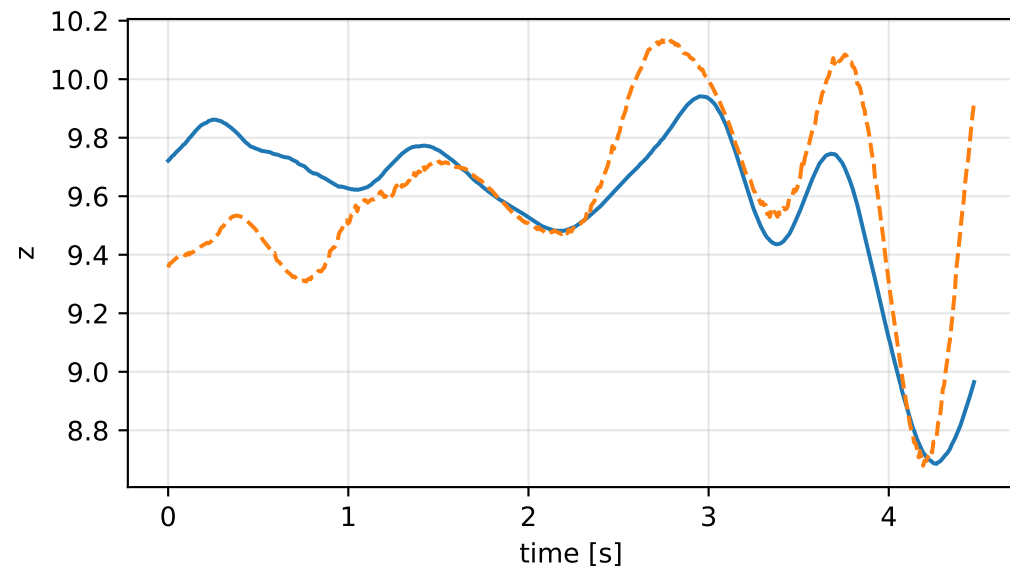
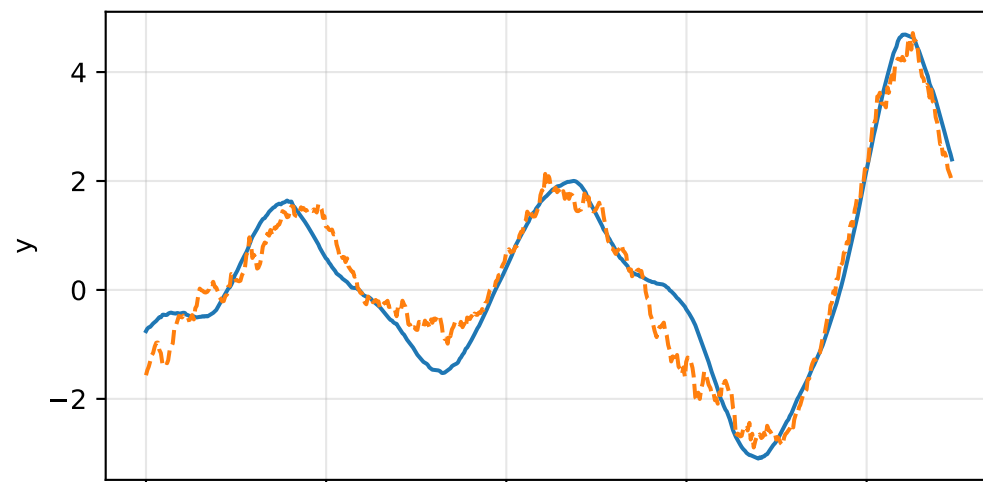
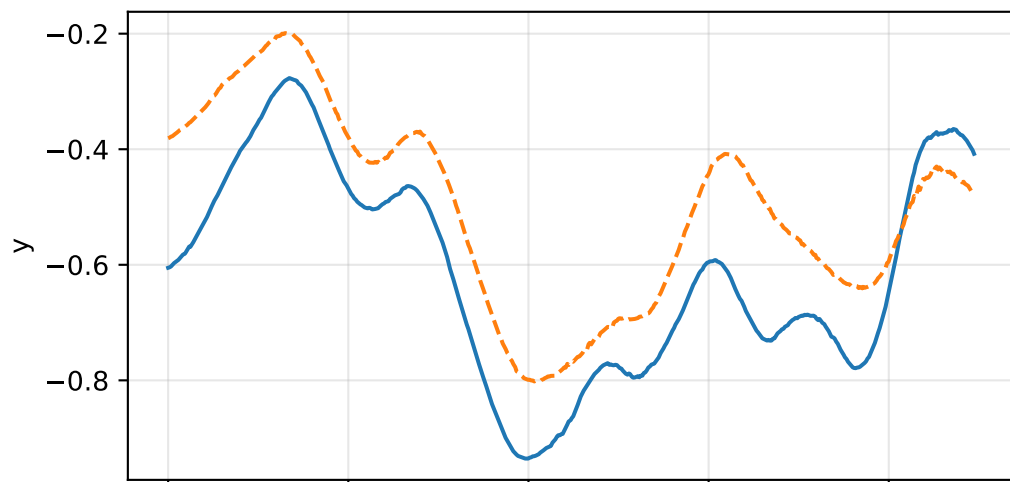
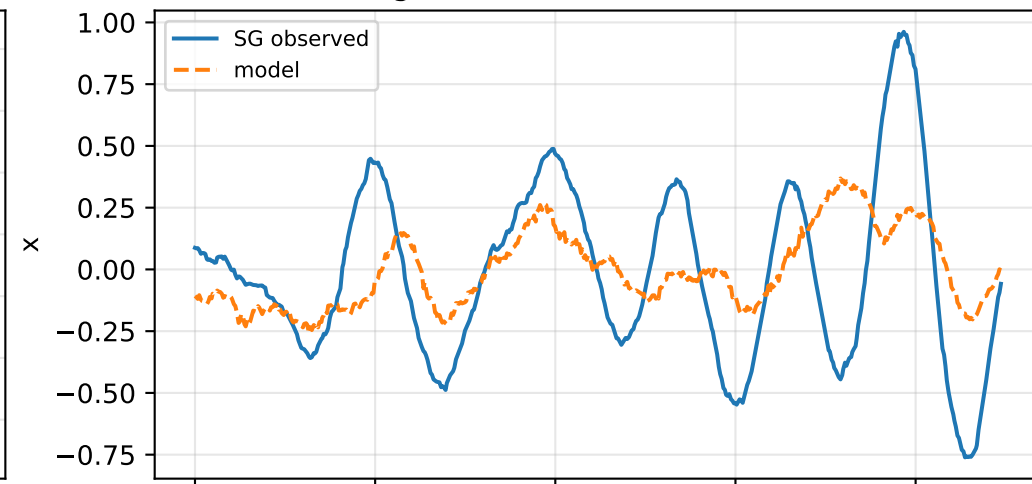


prior_free: acceleration residual objective

specific acceleration [m/s²]

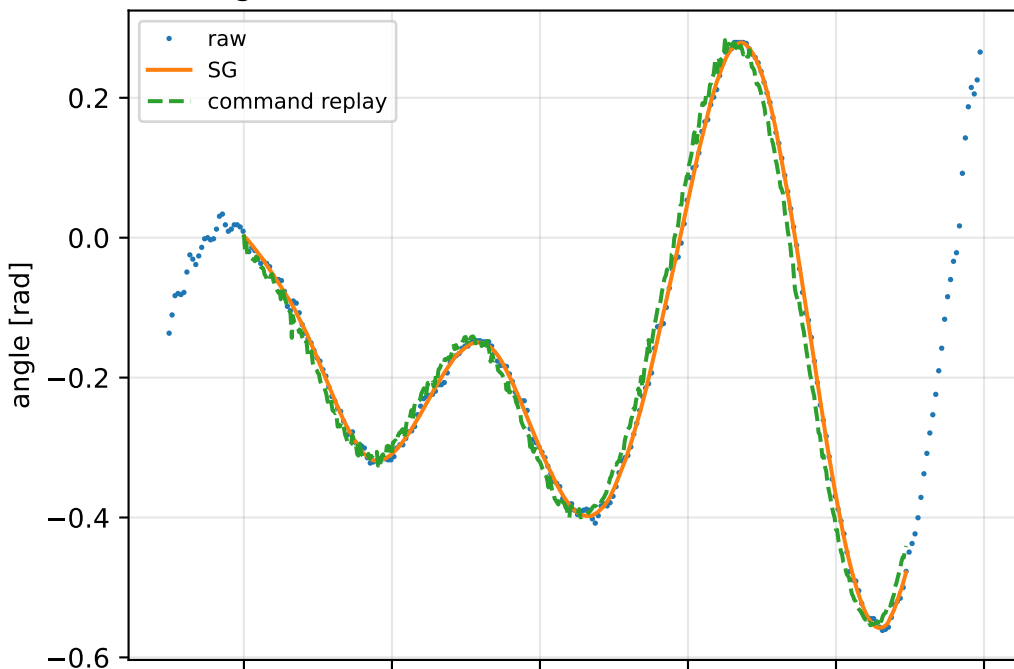


angular acceleration [rad/s²]

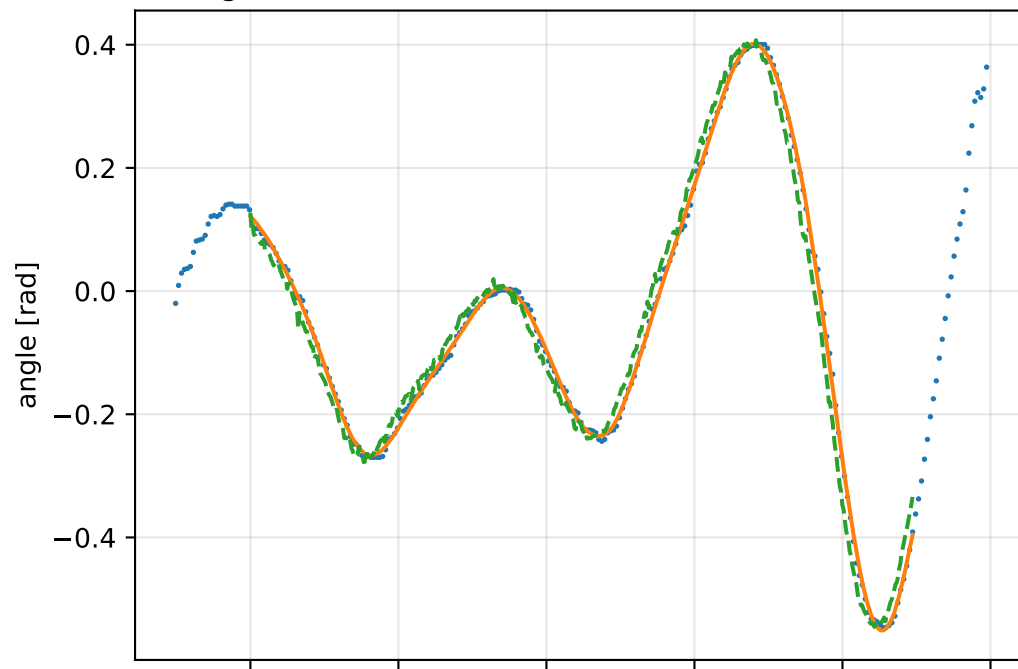


prior_free: gimbal measurement audit (objective=measured_sg)

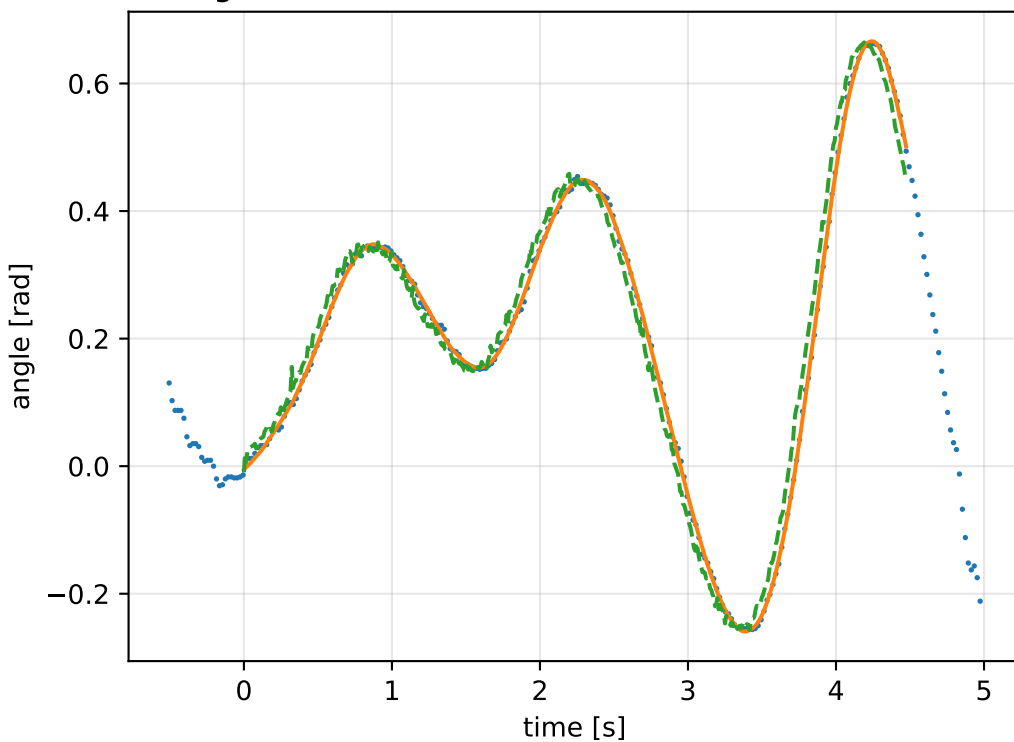
gimbal 1: RMSE=0.027, max=0.07529 rad



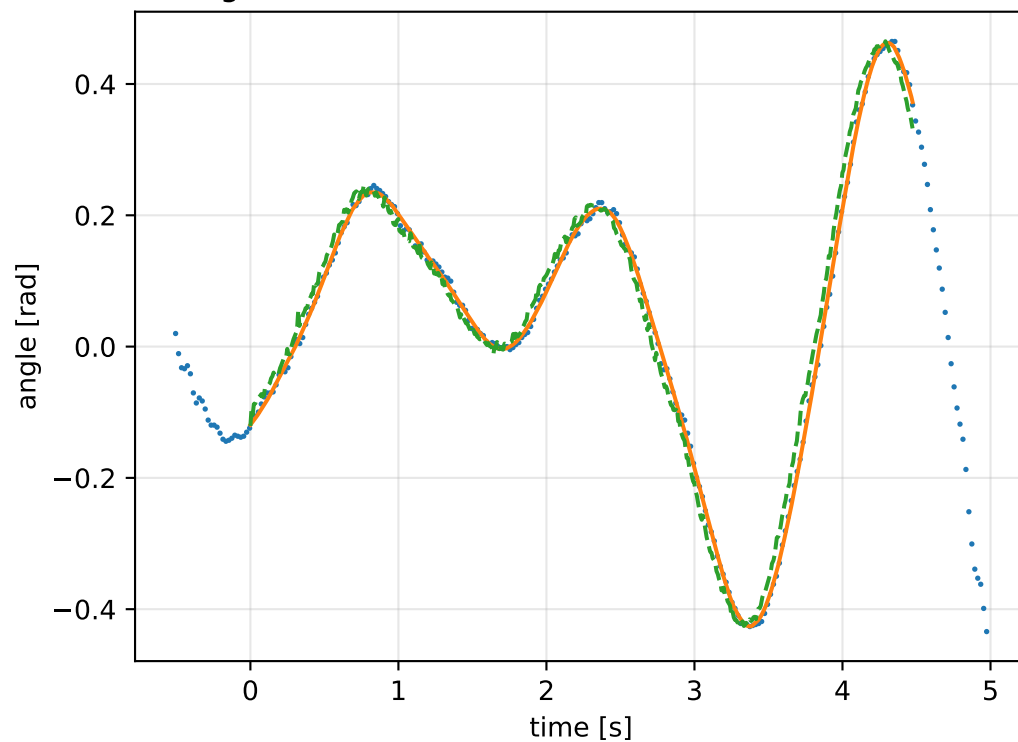
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

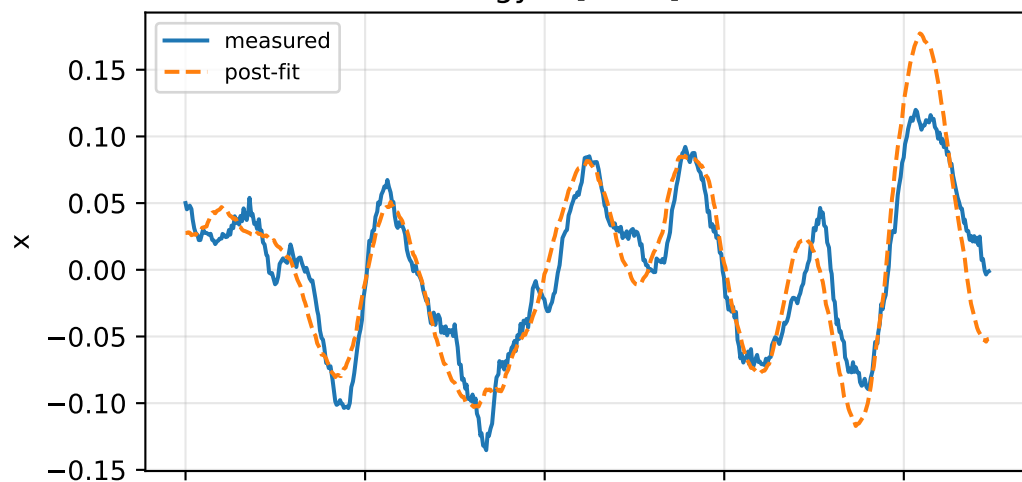


gimbal 4: RMSE=0.02598, max=0.07261 rad

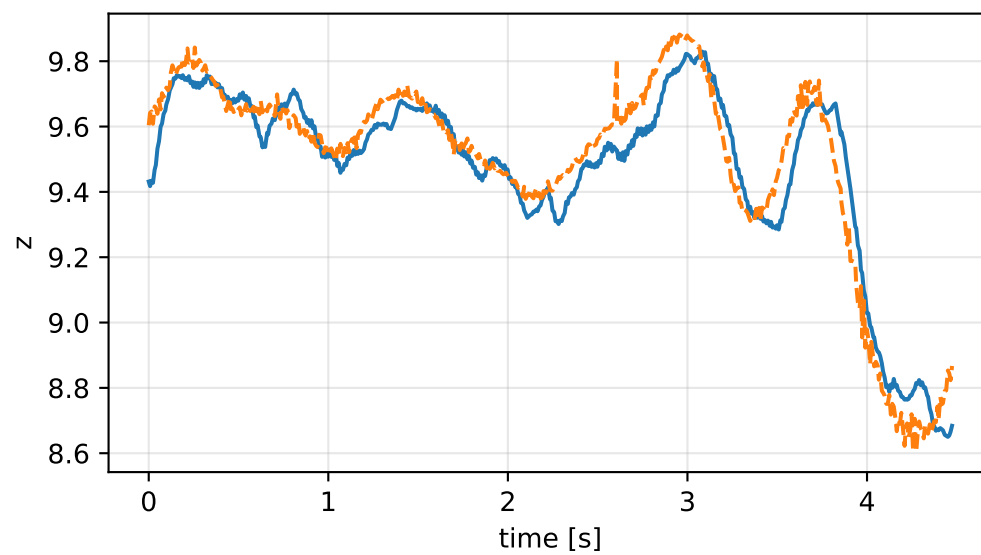
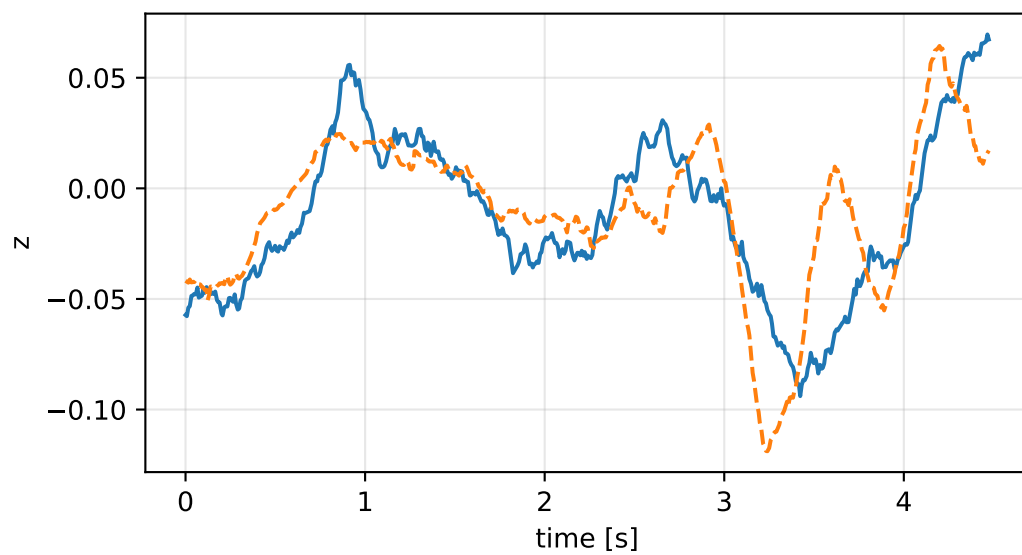
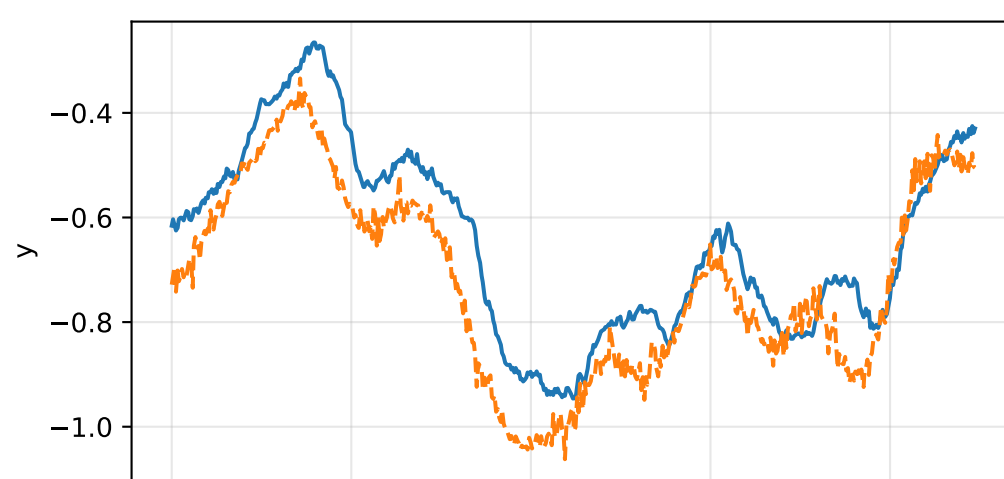
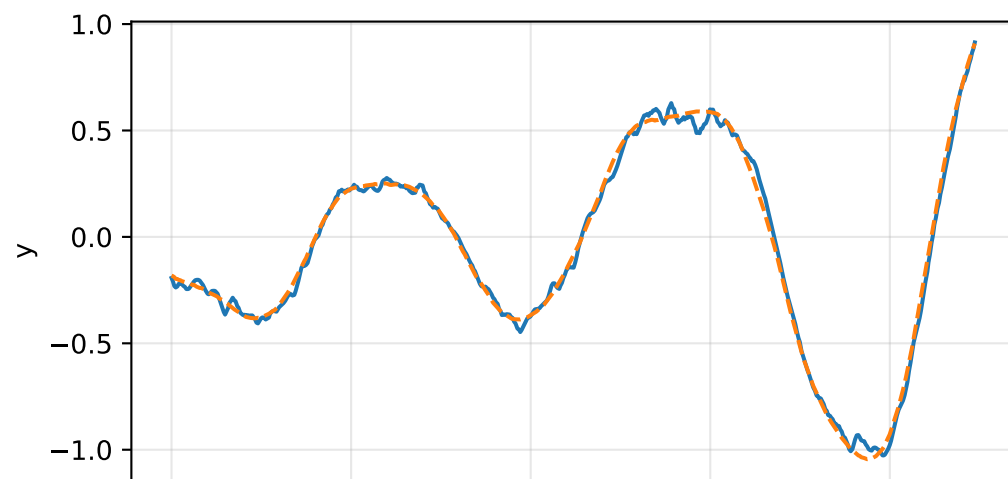
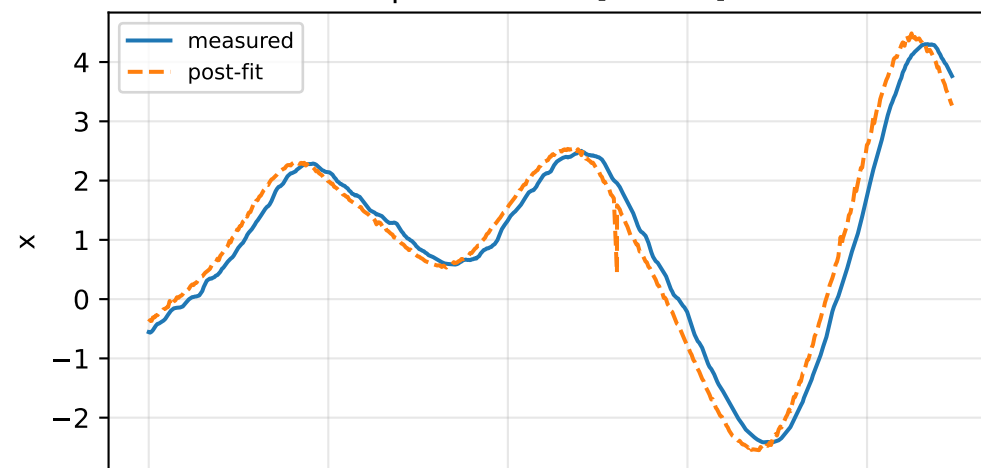


prior_free: IMU comparison (diagnostic only)

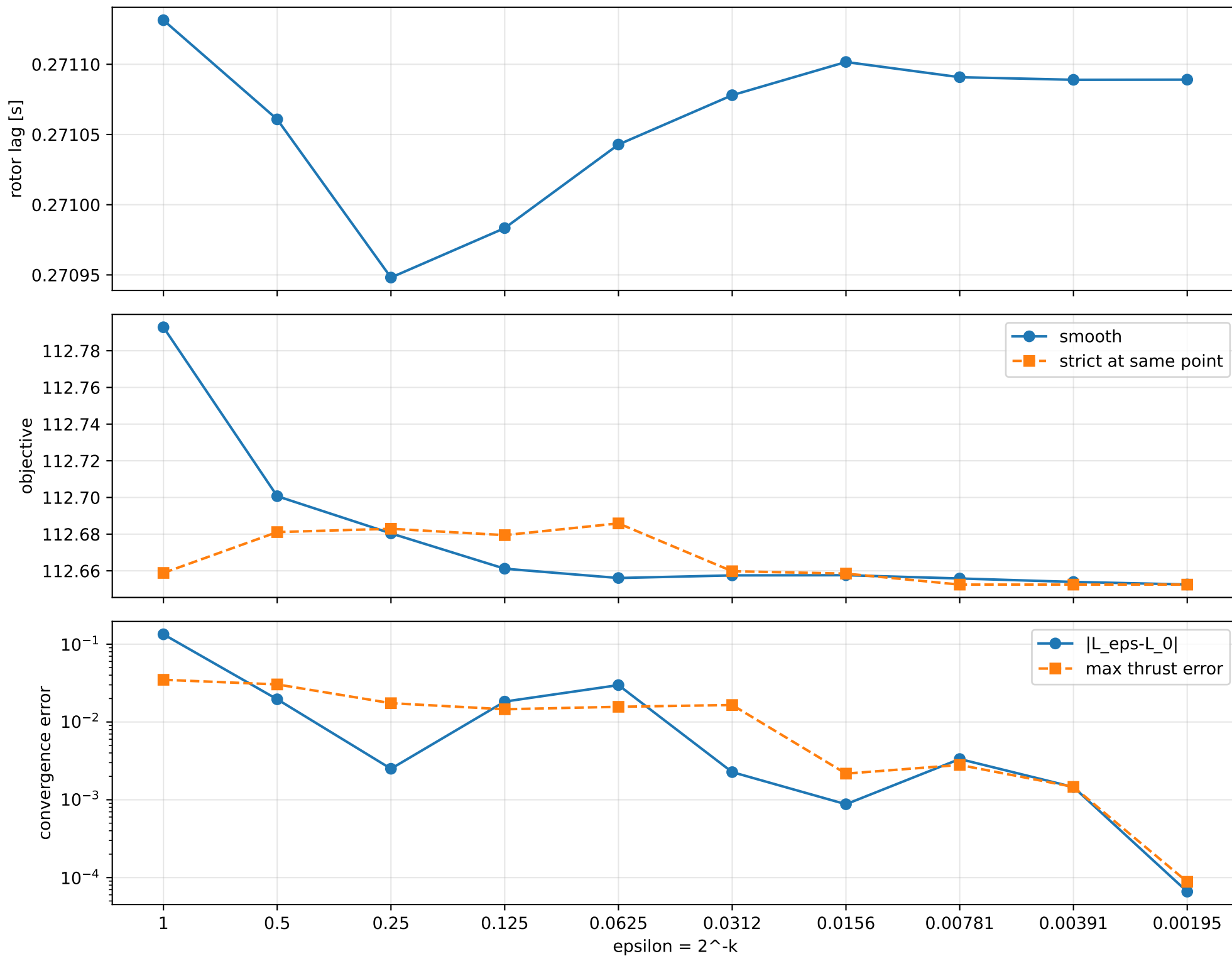
gyro [rad/s]



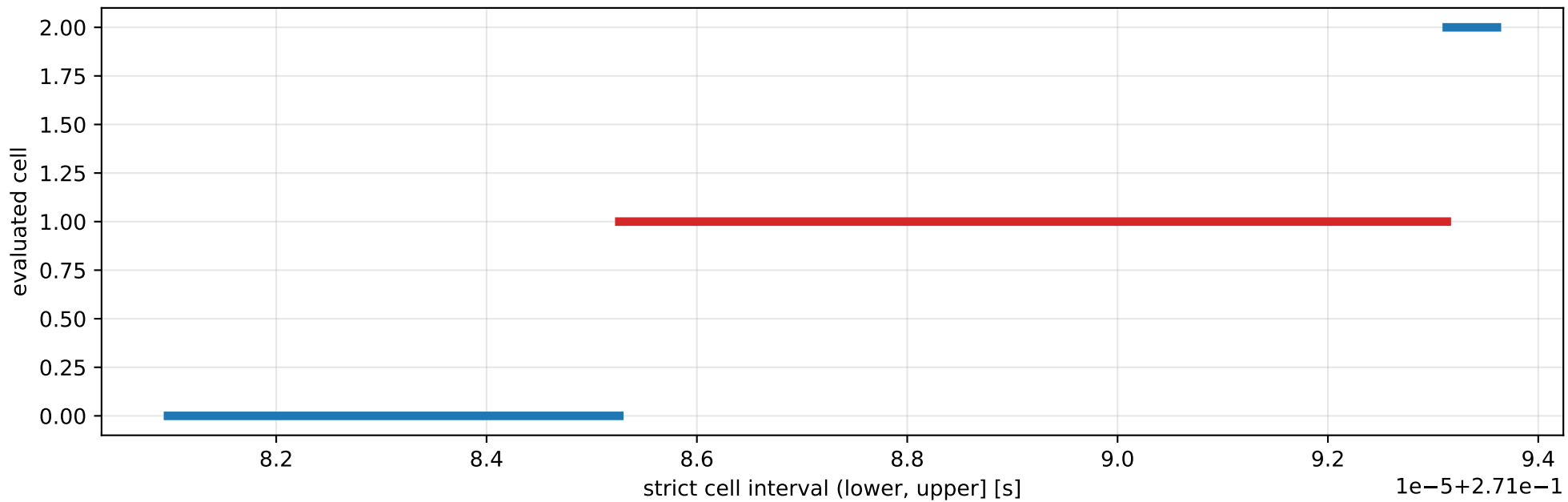
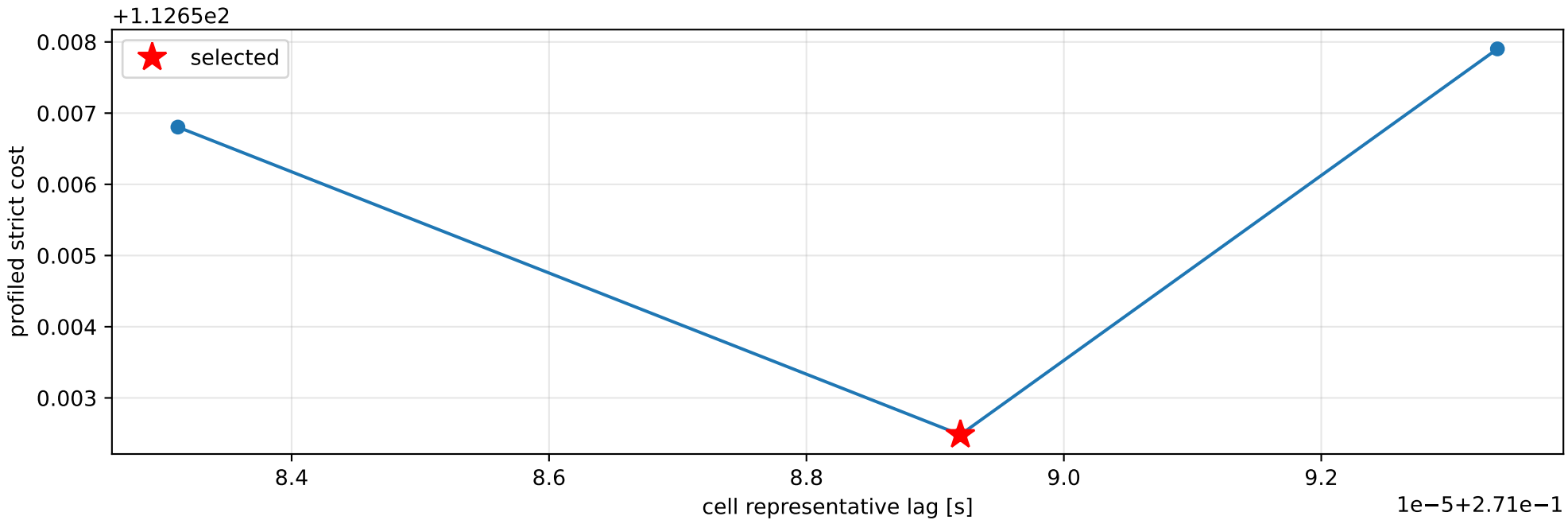
specific force [m/s^2]



prior_free: smooth-to-strict continuation



prior_free: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089061 s; data boundary=False

prior_free: parameter estimate

Case: prior_free

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.52043196

inertia [kg m²]:

```
[[ 0.55443708  0.01372501 -0.00323582]
 [ 0.01372501  0.26168335 -0.08172244]
 [-0.00323582 -0.08172244  0.79125005]]
```

CoG body [m]: [-0.026435 0.01133929 0.00941535]

force effectiveness: [2.8713137 2.32387853 2.56341481 3.07927651]

scale-free J/m [m²]:

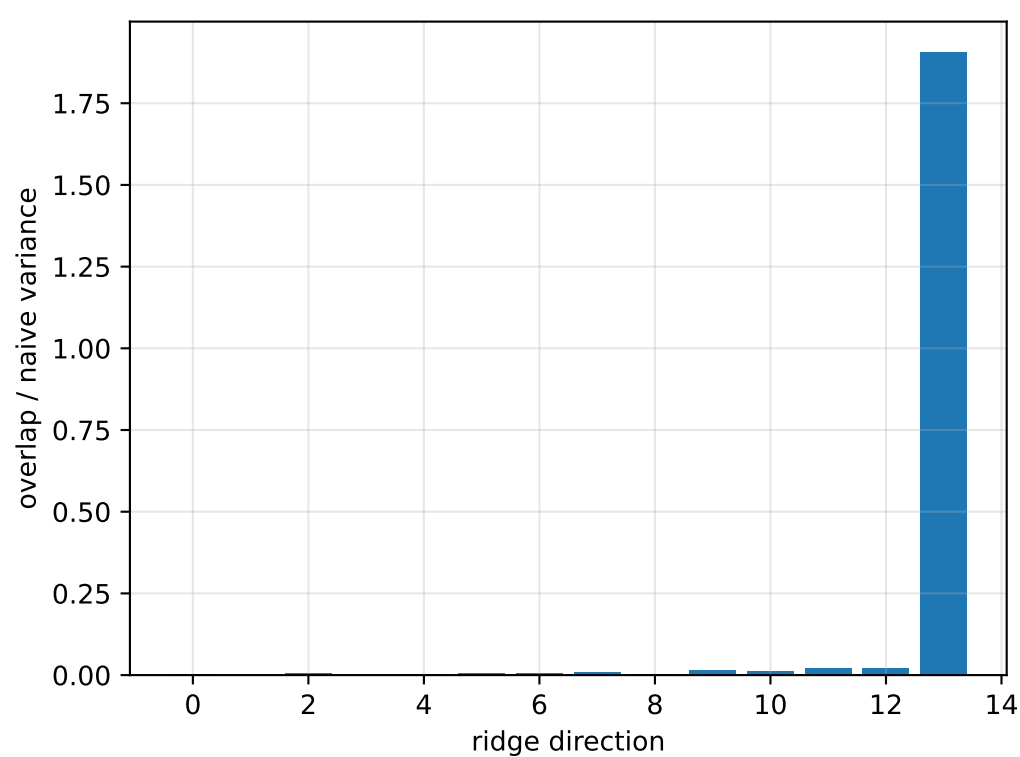
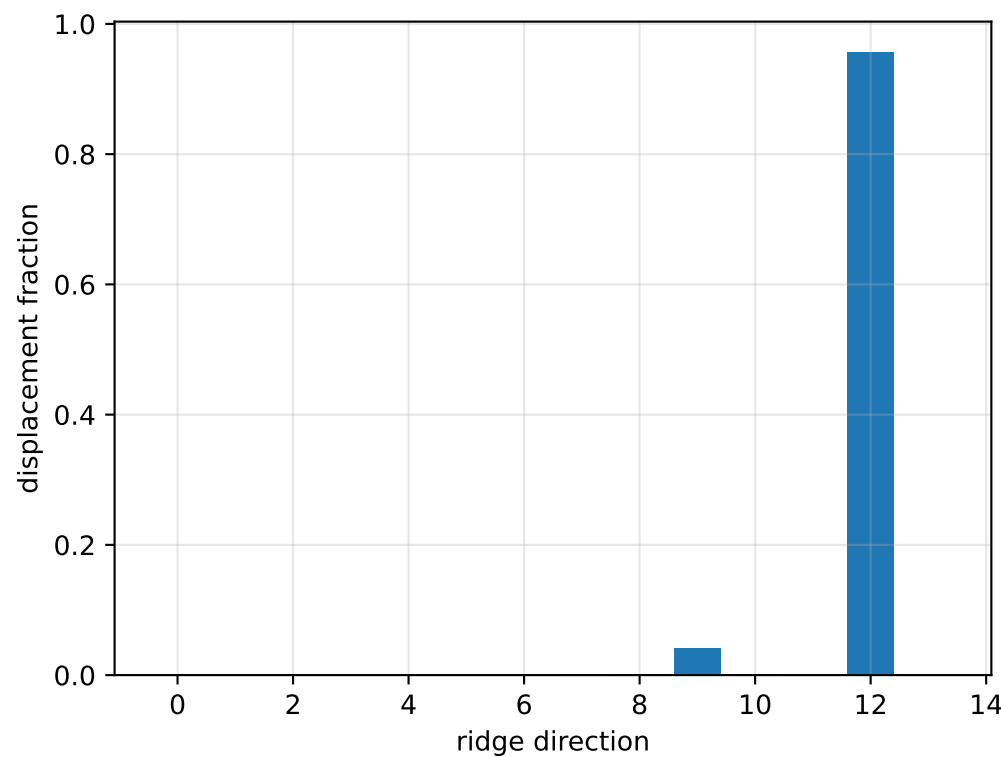
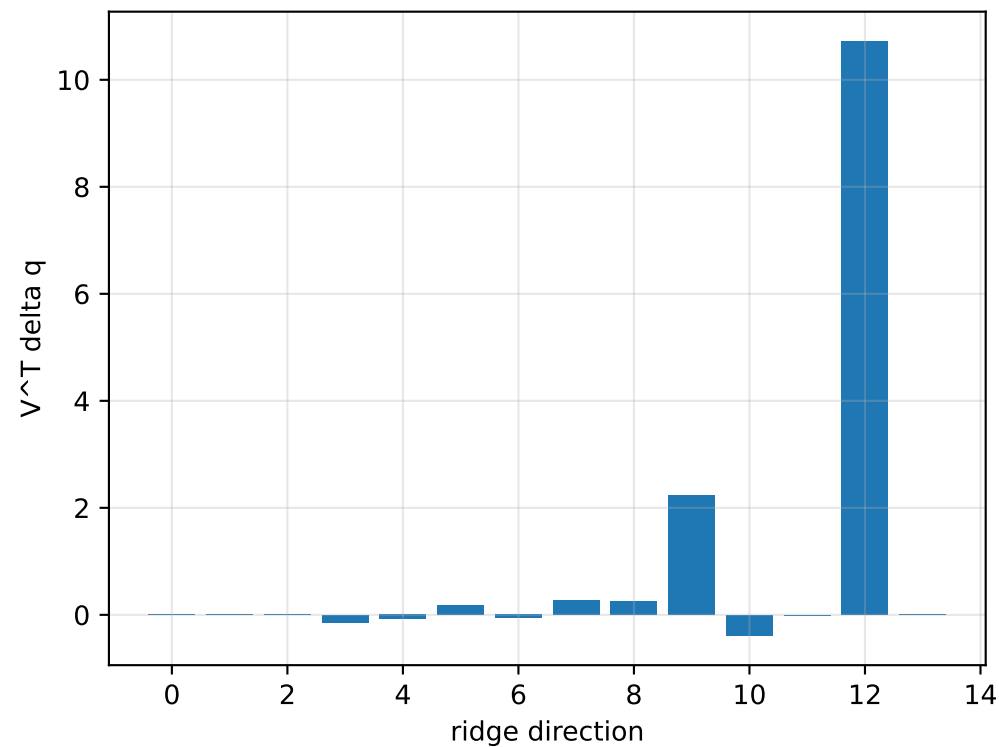
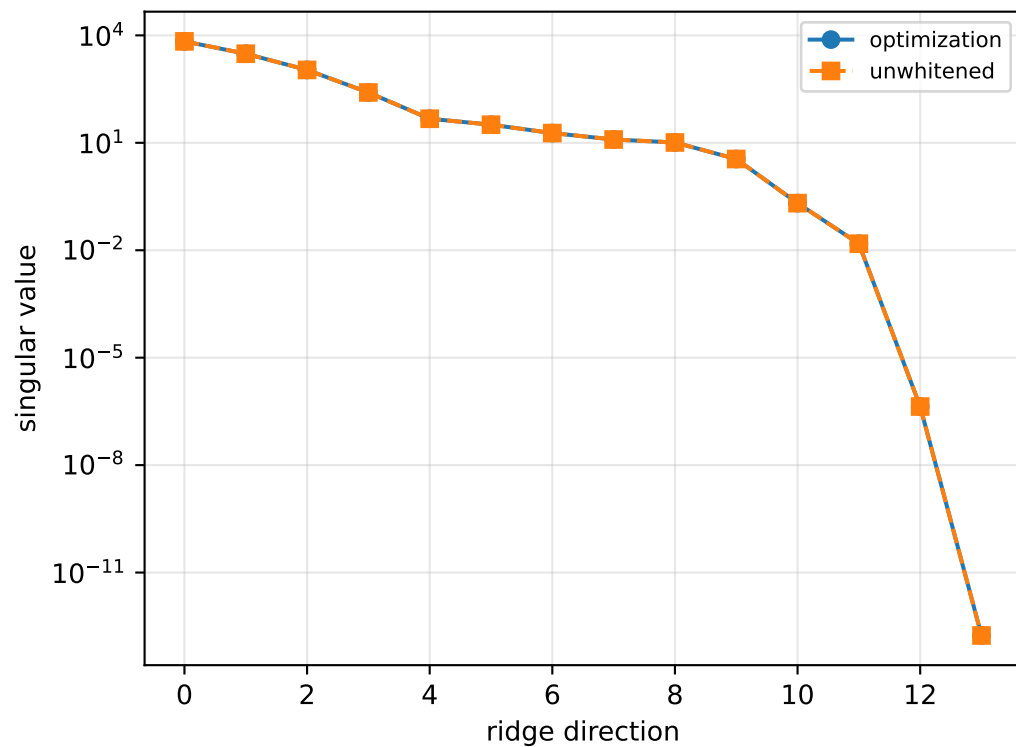
```
[[ 0.0737241  0.00182503 -0.00043027]
 [ 0.00182503  0.03479632 -0.01086672]
 [-0.00043027 -0.01086672  0.10521338]]
```

scale-free f/m: [0.3818017 0.30900865 0.34086005 0.40945474]

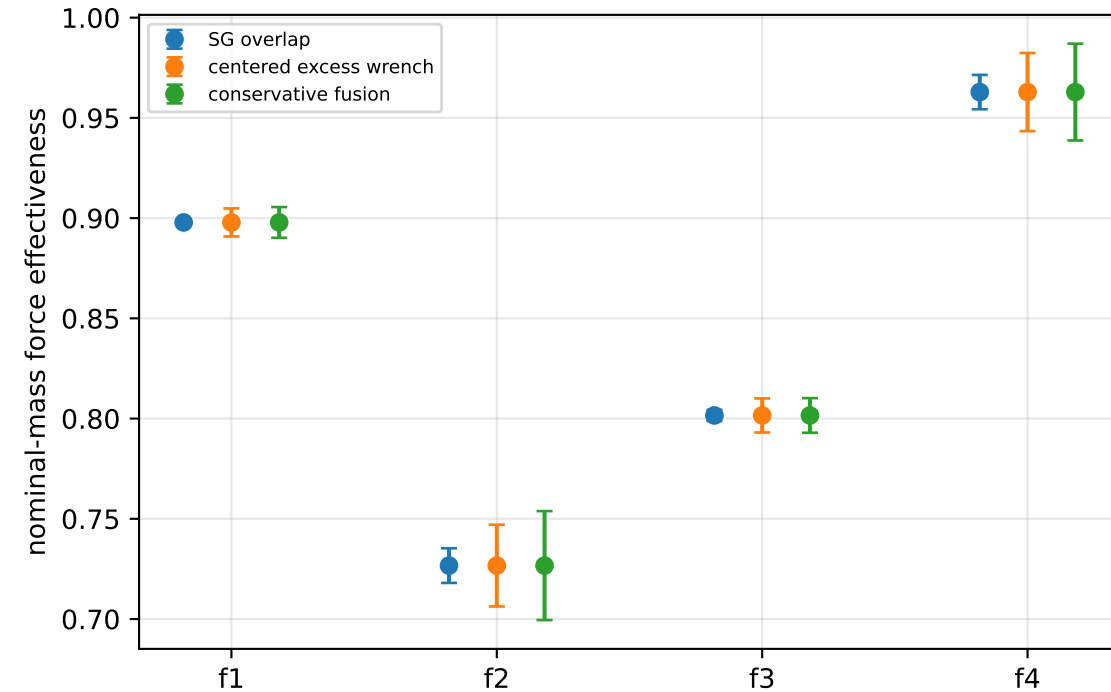
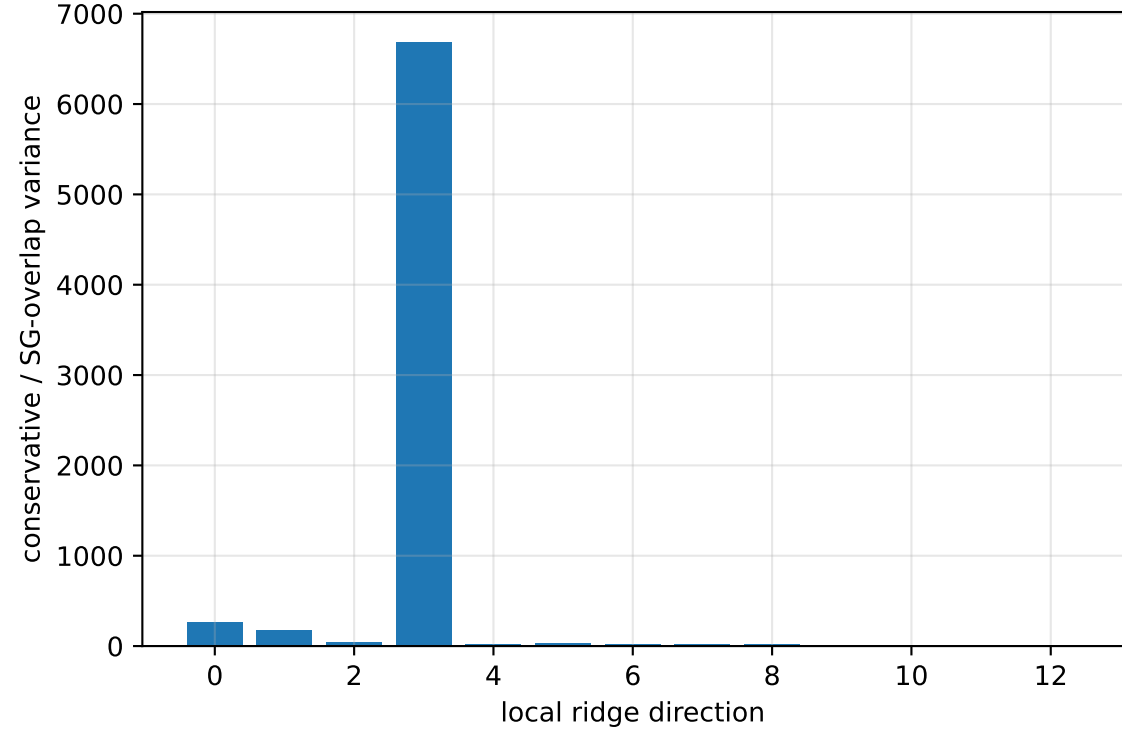
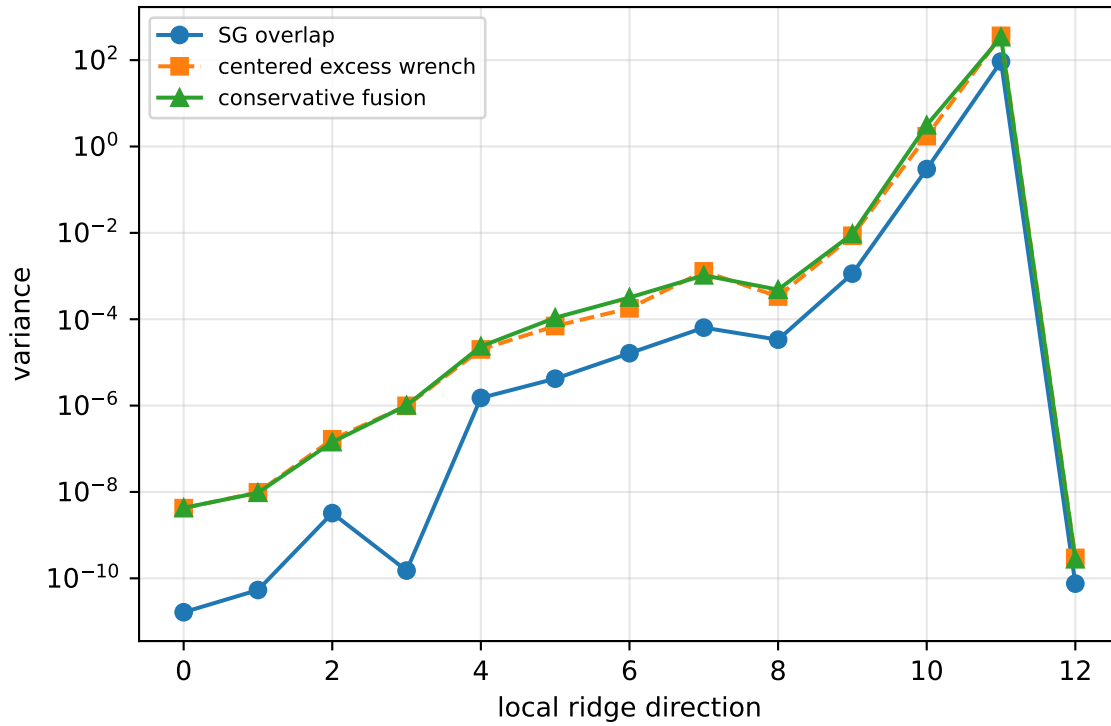
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

prior_free: ridge spectrum (rank 13, nullity 1, || $\mathbf{v}$ ||=1.63e-13)



prior_free: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 229.4$
wrench-to-acceleration recovery max error = $3.04\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

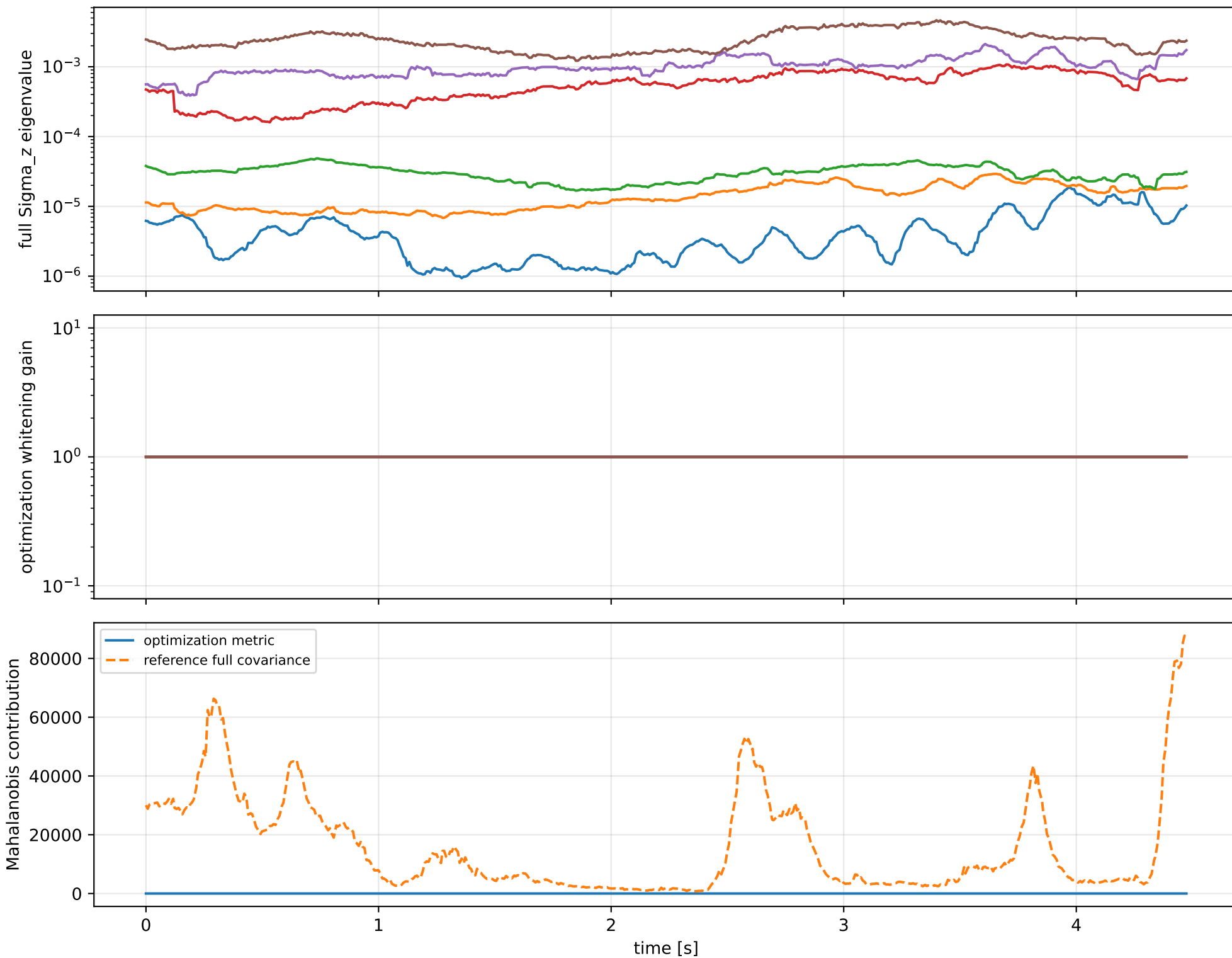
index 11: variance=340.5, ratio=3.668
 $\log_{\text{mass}}=+0.00219$; $\text{second_moment_diag_1}=+0.162$
 $\text{second_moment_diag_2}=-0.119$; $\text{second_moment_diag_3}=-0.0535$
 $\text{second_moment_offdiag_12}=+0.822$; $\text{second_moment_offdiag_13}=-0.517$
 $\text{second_moment_offdiag_23}=+0.114$; $\text{cog_x}=-5.2\text{e-}05$
 $\text{cog_y}=+3.09\text{e-}05$; $\text{cog_z}=-7.22\text{e-}06$
 $\log_{\text{force_eff_1}}=+0.00226$; $\log_{\text{force_eff_2}}=+0.00175$
 $\log_{\text{force_eff_3}}=+0.00208$; $\log_{\text{force_eff_4}}=+0.00251$

index 10: variance=3.11, ratio=10.36
 $\log_{\text{mass}}=+0.0249$; $\text{second_moment_diag_1}=+0.0251$
 $\text{second_moment_diag_2}=+0.561$; $\text{second_moment_diag_3}=-0.713$
 $\text{second_moment_offdiag_12}=+0.0341$; $\text{second_moment_offdiag_13}=+0.0961$
 $\text{second_moment_offdiag_23}=+0.403$; $\text{cog_x}=+0.000943$
 $\text{cog_y}=-0.00131$; $\text{cog_z}=+0.000291$
 $\log_{\text{force_eff_1}}=+0.0256$; $\log_{\text{force_eff_2}}=+0.0377$
 $\log_{\text{force_eff_3}}=+0.0224$; $\log_{\text{force_eff_4}}=+0.0172$

index 9: variance=0.009426, ratio=8.286
 $\log_{\text{mass}}=-0.141$; $\text{second_moment_diag_1}=-0.132$
 $\text{second_moment_diag_2}=+0.323$; $\text{second_moment_diag_3}=+0.567$
 $\text{second_moment_offdiag_12}=-0.00826$; $\text{second_moment_offdiag_13}=-0.0513$
 $\text{second_moment_offdiag_23}=+0.631$; $\text{cog_x}=-0.0198$
 $\text{cog_y}=+0.0106$; $\text{cog_z}=-0.00278$
 $\log_{\text{force_eff_1}}=-0.104$; $\log_{\text{force_eff_2}}=-0.3$
 $\log_{\text{force_eff_3}}=-0.188$; $\log_{\text{force_eff_4}}=-0.0253$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

prior_free: optimization vs reference covariance



prior_free: wrench / closure (force max 2.29e-14, torque max 4.79e-16)

